LESSON 7.3

Understand Absolute Value of Rational Numbers

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find the absolute value of a rational number — an integer, fraction, or decimal — as its distance from zero.

WHY THIS MATTERS Claire is performing a science experiment with two liquids. The temperatures of the two liquids in Celsius are shown. She needs to use the beaker with the temperature that is closer to 0°C. Which beaker should Claire use?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Integer *Spanish: Número entero*	Whole numbers and their opposites, like -2, -1, 0, 1, 2.	... -3, -2, -1, 0, 1, 2, 3 ... shown on a number line with 0 in the center
Positive *Spanish: Positivo*	A number bigger than zero, to the right of zero on a number line.	+5 means 5 above zero, like 5 floors above ground level
Negative *Spanish: Negativo*	A number smaller than zero, to the left of zero on a number line.	-3 means 3 below zero, like 3 floors underground
Absolute value *Spanish: Valor absoluto*	How far a number is from zero. It is never negative.	$\|-3\| = 3$ and $\|3\| = 3$
Opposite *Spanish: Opuesto*	Two numbers the same distance from zero, on opposite sides, like 3 and -3.	−3 and 3 are opposites
Rational number *Spanish: Número racional*	A number that can be written as a fraction of two integers, with a bottom number that is not zero.	−3/4, 0.5, 7, −50

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Find $|-13|$ and $|6|$. Which number is farther from 0?

1. −13 is 13 units left of 0, so $|-13| = 13$.
2. 6 is 6 units right of 0, so $|6| = 6$.
3. Compare the distances: $13 > 6$.

Answer: −13 is farther from 0

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Now let's find |3|. Where is 3 on the number line? It is 3 steps to the right of zero.

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The absolute value of _____ is _____ because it sits _____ units from zero.
- _____ and _____ share the same absolute value because _____ .
- An absolute value is never negative because _____ .
- The number closer to zero is _____ because its distance is _____ .

Word bank absolute value | distance | magnitude | zero | number line | opposite | units | closer | farther | negative | positive | sign

WATCH OUT!

A common mistake

A common mistake in Integers and Absolute Value is judging size by the digits alone instead of position on the number line

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Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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1 **Circle the letter of the best answer. Show how you know.**

What is the absolute value of -9?

(A) -9 (B) 0

(C) 1 (D) 9

STEPS TO TRY

1. Absolute value asks: how FAR is -9 from zero? It's about distance, not direction.
2. Picture -9 on a number line and count the hops back to 0. A distance is never negative.
3. Start at -9 and count the steps to reach 0: 1, 2, 3, ... How many hops does it take in all?

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Which integer has the GREATEST absolute value: -3, 7, -5, 2?

(A) -5
(B) -3
(C) 2
(D) 7

STEPS TO TRY

1. 'Greatest absolute value' means farthest from zero — the sign doesn't matter.
2. Find each number's distance from 0: |-3|, |7|, |-5|, |2|. Then pick the biggest distance.
3. Start with |-3| = 3 and |-5| = 5. Now find the distances for 7 and 2 and compare all four.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

What is |0|?

(A) 1
(B) -1
(C) 0
(D) There is no answer

STEPS TO TRY

1. Absolute value = distance from zero. Here the number inside the bars IS zero.
2. Ask: how many units do you travel from 0 to reach 0 on the number line?
3. Place your finger on 0. Count the hops needed to get to 0 — that count is the value of |0|.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

What is |−2.5|?

(A) 0

(B) 2.5

(C) −2.5

(D) 25

STEPS TO TRY

1. Where does −2.5 sit on the number line?
2. How far is that from zero?
3. Distance ignores direction.

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How does the number line help you see the difference between an integer and its absolute value?

SENTENCE STARTER The integer ___ is located ___ units from zero, so its absolute value is ___. Even though ___ is negative, its distance from zero is still ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help